Cooling system and power management

Q-1 Measure your CPU temperature at idle and under load using a tool like HWMonitor or Core Temp, then record the readings before and after applying new thermal paste and reinstalling your CPU cooler.

ANS: **CPU Temperature Test**

Tool Used: HWMonitor / Core Temp
Test: CPU temperature before and after applying new thermal paste and reinstalling the CPU cooler.

Condition	Before Thermal Paste	After Thermal Paste
Idle	___ °C	___ °C
Under Load	___ °C	___ °C

Procedure

1. Install and open **HWMonitor** or **Core Temp**.
2. Start the PC and leave it idle for about **10–15 minutes**.
3. Record the CPU temperature under **Idle**.
4. Run a CPU stress test or a heavy application for about **10–15 minutes**.
5. Record the CPU temperature under **Load**.
6. Shut down the PC and unplug it.
7. Remove the CPU cooler carefully.
8. Clean the old thermal paste from the CPU and cooler.
9. Apply a small amount of new thermal paste.
10. Reinstall the CPU cooler properly.
11. Repeat the same idle and load tests.
12. Record the new temperatures and compare them.

Q- 2 Write a step-by-step guide (with photos or diagrams) showing how to correctly apply thermal paste and mount a CPU cooler for optimal heat transfer.

Hint: Include tips on how much paste to use, spreading methods, and common mistakes to avoid.

ANS: Real-World Storage Recommendations

Scenario	Recommended Storage	Suggested Capacity	Reason
1. Budget Gaming PC	500GB SATA SSD	500GB	A SATA SSD gives faster game loading and system startup than an HDD while keeping the cost lower than a high-end NVMe SSD. 500GB is a reasonable starting capacity for a budget gaming PC.
2. College Student Laptop – Coding + Netflix	500GB NVMe SSD	500GB	An NVMe SSD provides fast application launching, quick boot times, and smooth everyday use. 500GB gives enough space for coding tools, projects, documents, and normal media use.
3. High-End Video Editing Workstation	1TB or 2TB NVMe SSD + large HDD	1–2TB SSD + 2–4TB HDD	NVMe SSD is suitable for active video projects, editing and fast file access, while a large HDD can be used for storing raw footage and completed projects at a lower cost per GB.

Q-3 Calculate the recommended PSU wattage for three different PC builds: a basic office PC, a gaming PC, and a content creator PC (with GPU and multiple drives). List all major components and show your calculations for each build.

ANS: **PSU Wattage Calculation for Three PC Builds**

Formula

Recommended PSU = Total Component Power × 1.30

The extra 30% provides some power headroom for normal power spikes and future upgrades.

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1. Basic Office PC

Major Components

Component	Estimated Power
CPU – Intel Core i3	60 W
Motherboard	50 W
RAM – 16 GB	10 W
SSD – 500 GB	5 W
Case Fans	10 W
USB/Other Devices	15 W
Total	**150 W**

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2. Gaming PC

Major Components

Component	Estimated Power
CPU – Intel Core i5-12400F	117 W
GPU – NVIDIA RTX 4060	115 W
Motherboard	50 W
RAM – 16 GB	10 W
NVMe SSD – 1 TB	7 W
Case Fans	15 W
USB/Other Devices	15 W
Total	**329 W**

Intel specifies up to 117 W maximum turbo power for the i5-12400F, and NVIDIA lists 115 W total graphics power for the RTX 4060.

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3. High-End Content Creator PC

Major Components

Component	Estimated Power
CPU – AMD Ryzen 9 7900X	170 W
GPU – High-end GPU	300 W
Motherboard	70 W
RAM – 64 GB	20 W
NVMe SSD – 2 TB	10 W
SATA SSD – 2 TB	5 W
HDD – 4 TB	10 W
Case/CPU Fans	30 W
USB/Other Devices	20 W
Total	**635 W**

Final Comparison

PC Build	Estimated Total Power	30% Headroom	Recommended PSU
Basic Office PC	150 W	195 W	**300 W**
Gaming PC	329 W	428 W	**550 W**
Content Creator PC	635 W	826 W	**850 W**

Q-4 Compare the total 5-year electricity cost for two PSUs: one rated 80 Plus Bronze (85% efficiency) and one rated 80 Plus Gold (90% efficiency), both running a 400W average load for 8 hours daily at ₹8 per kWh. Show your math and final cost difference.

Hint: Remember to account for efficiency when calculating actual power drawn from the wall.

ANS: **5-Year Electricity Cost: Bronze vs Gold PSU**

Given Information

- Average PC load = **400 W**
- Usage = **8 hours per day**
- Electricity rate = **₹8 per kWh**
- Time period = **5 years**
- 80 Plus Bronze efficiency = **85%**
- 80 Plus Gold efficiency = **90%**

1. 80 Plus Bronze PSU

Step 1: Calculate power drawn from the wall

Wall Power = PC Load ÷ Efficiency

= 400 ÷ 0.85
= **470.59 W**

Step 2: Daily electricity usage

470.59 W × 8 hours = 3764.72 Wh

= **3.7647 kWh/day**

Step 3: 5-year electricity usage

3.7647 × 365 × 5
= **6,871.6 kWh**

Step 4: Total cost

6,871.6 × ₹8
= **₹54,972.80**

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2. 80 Plus Gold PSU

Step 1: Calculate power drawn from the wall

Wall Power = 400 ÷ 0.90

= **444.44 W**

Step 2: Daily electricity usage

444.44 W × 8 hours = 3555.52 Wh

= **3.5556 kWh/day**

Step 3: 5-year electricity usage

3.5556 × 365 × 5
= **6,488.9 kWh**

Step 4: Total cost

6,488.9 × ₹8
= **₹51,911.20**

Final Comparison

PSU	Efficiency	Wall Power	5-Year Electricity Cost
80 Plus Bronze	85%	470.59 W	**₹54,972.80**
80 Plus Gold	90%	444.44 W	**₹51,911.20**

Cost Difference

₹54,972.80 – ₹51,911.20 = ₹3,061.60

Therefore, the **80 Plus Gold PSU would save approximately ₹3,062 in electricity costs over 5 years** under these assumptions.

Q-5 Create a table comparing air cooling vs liquid cooling for desktop PCs, listing at least 3 pros and 3 cons for each based on performance, noise, maintenance, and cost.

ANS: **Air Cooling vs Liquid Cooling**

Point	Air Cooling	Liquid Cooling
Performance – Pro 1	Good cooling for normal office and gaming PCs.	Very effective for high-performance CPUs and heavy workloads.
Performance – Pro 2	Large heatsinks can provide strong cooling without a pump.	Radiators can move heat away from the CPU efficiently.
Performance – Pro 3	Works well with good case airflow.	Useful for high-power CPUs and overclocking.
Performance – Con	Very large air coolers may struggle with extremely high heat loads.	AIO/custom systems are more complex and require more components.
Noise – Pro	Modern air coolers can use quiet fans at lower speeds.	Can be quieter under heavy loads because radiator fans may run at lower RPM.

Point	Air Cooling	Liquid Cooling
Noise – Con	Fans can become louder when CPU temperature increases.	Pump and radiator fans can produce noise.
Maintenance – Pro	Simple design with no pump or coolant.	AIO liquid coolers normally require little maintenance after installation.
Maintenance – Con	Dust must be cleaned from the heatsink and fans regularly.	More parts mean more possible points of failure; custom loops require much more maintenance.
Cost – Pro	Usually cheaper and offers good value.	Premium liquid coolers can provide strong cooling for high-end systems.
Cost – Con	Large premium air coolers can still be expensive.	Usually costs more than a basic air cooler.
Installation	Usually simple and easy to install.	AIO installation is more complicated because the radiator also needs to be mounted.
Space	A large heatsink can take up space around the CPU and RAM.	The CPU area can be less crowded, but the radiator needs suitable case space.